

# Merge K sorted Linked List



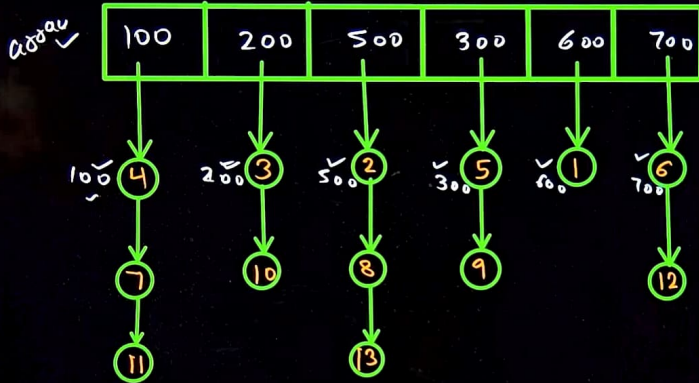
$K=6$

① Brute force

② Merge-sort

$nk \log k$

# Merge K sorted Linked List



K=6

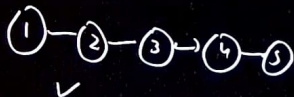
① Brute force

② Merge-sort

$\frac{n \log k}{k}$

③  $\Rightarrow$  Heap ✓

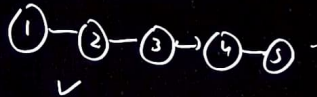
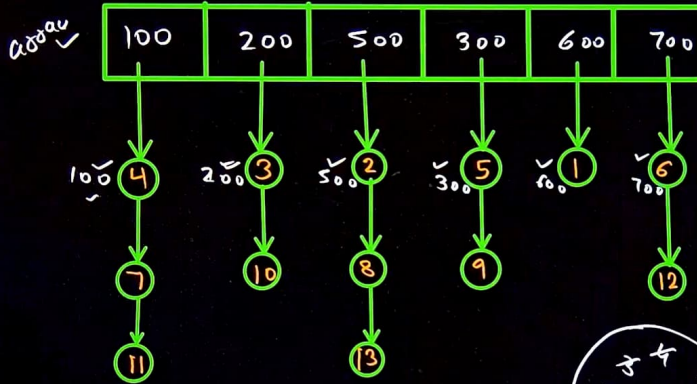
01 2



③  $\Rightarrow$  Heap  $\checkmark$



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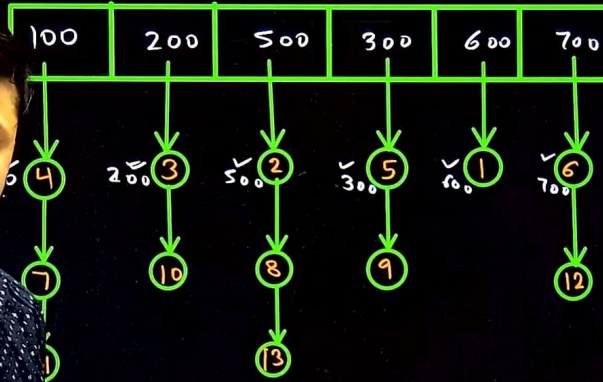
$\frac{nK \log K}{}$

③ ⇒ Heap ✓

$\left[ \frac{nK \log nK}{nK \log nK} \right]$   
 $\boxed{nK \log nK}$



# Merge K sorted Linked List



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② Merge-sort

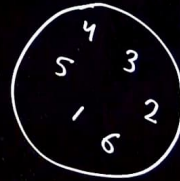
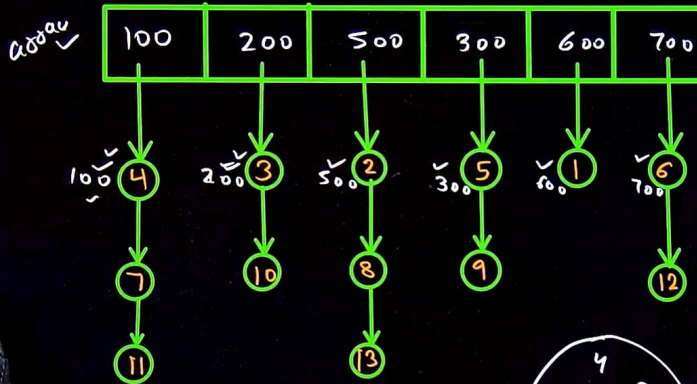
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③  $\Rightarrow$  Heap  $\checkmark$   
space  $\rightarrow nk$

$\left[ \frac{nk \log nk}{+ nk \log nk} \right]$   
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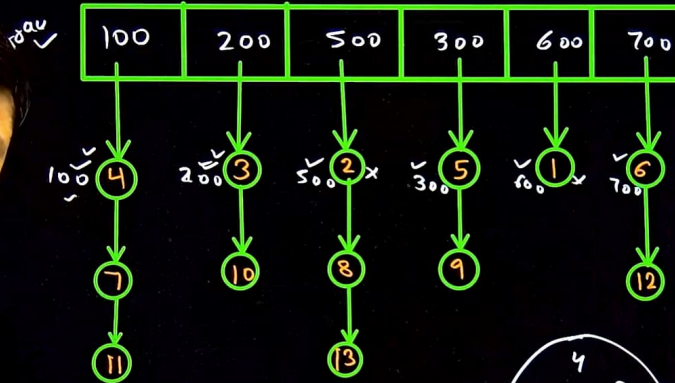
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# Merge K sorted Linked List



① - ②



K=6

- ① Brute force
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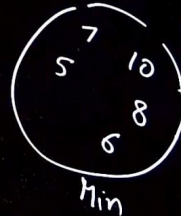
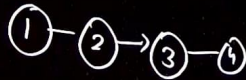
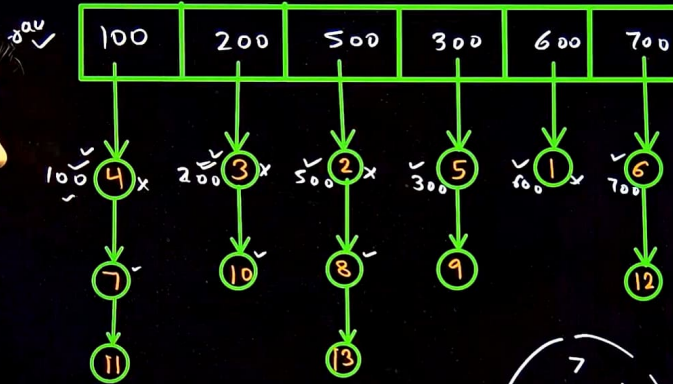
$\frac{nk \log k}{}$

③  $\Rightarrow$  Heap  $\checkmark$   
Space  $\rightarrow nk$

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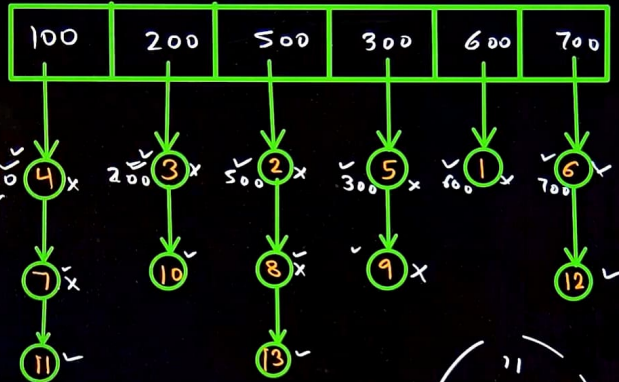
③ ⇒ Heap ✓  
space → nk

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$\frac{nk \log nk}{}$



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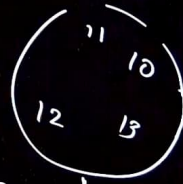
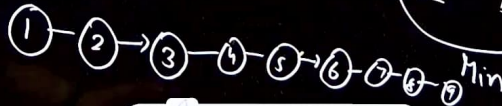
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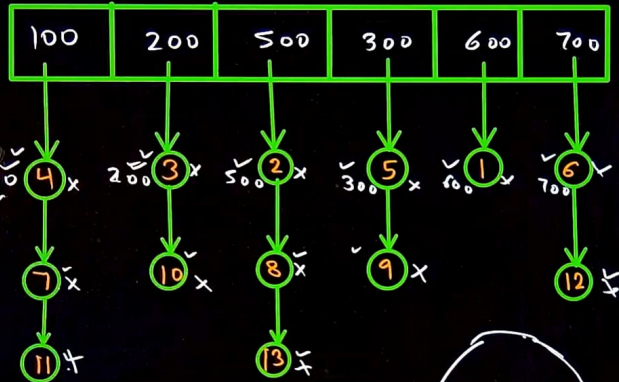
$$\left[ \frac{nk \log nk}{nk \log nk} \right]$$

$\boxed{nk \log nk}$



# Merge K sorted Linked List

K-elements:



$$\left( \log K + \log K + 1 \right) \Rightarrow \log K$$

$K=6$

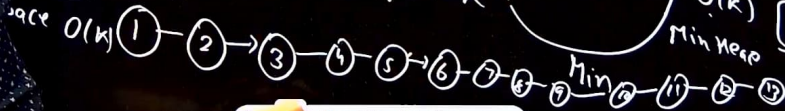
- ① Brute force
- ② Merge-sort

$$\frac{nK \log K}{\log K}$$

③  $\Rightarrow$  Heap  $\checkmark$   
Space  $\rightarrow nK$

$$\left[ \frac{nK \log nK}{nK \log nK} \right] = nK \log nK$$

$$[nK \log K] \rightarrow (n-K) \log K + K$$



$O(K)$   
Min Heap

# Merge K sorted Linked List

27/05/2024

K-elements:

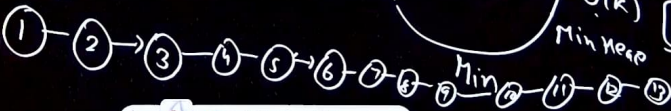


$\log k$

$\log k$

$[nk \log k]$

Space  $O(k)$



$$\left( \log k + \log k + 1 \right) \Rightarrow \log k$$

$K=6$

- ① Brute force
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$nk \log k$

③  $\Rightarrow$  Heap  $\checkmark$   
Space  $\rightarrow nk$

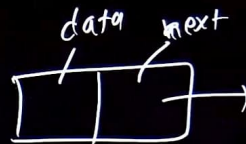
$$\left[ \frac{nk \log nk}{nk \log nk} \right]$$

$nk \log nk$



class Node

```
{ public:  
    int data;  
    Node * next;  
    Node (int x)  
    { data = x;  
      next = NULL;  
    }
```



Priority\_queue < int, vector<int>, greater<int>> > p;

# Merge K sorted Linked List

K-elements: root  $\rightarrow$  data  
 root  $\rightarrow$  next  $\rightarrow$  (2)  $\rightarrow$  next  $\rightarrow$  300

$$\left( \log K + \log K + 1 \right) \Rightarrow \log K$$

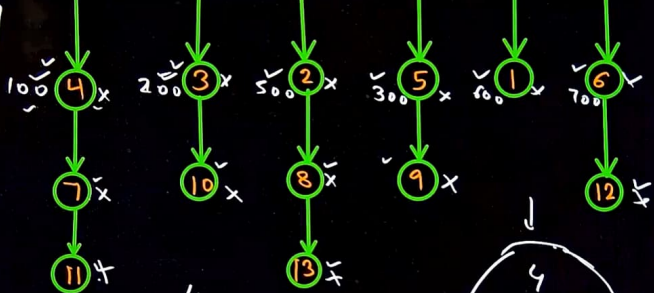
$K=6$

- ① Brute force
- ② Merge-sort

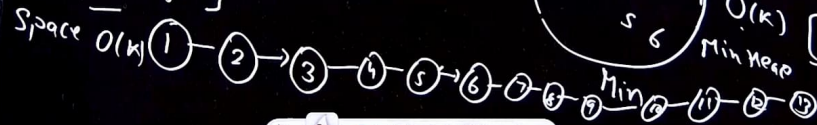
$$\frac{nk \log k}{nk \log k}$$

③  $\Rightarrow$  Heap  $\checkmark$   
 Space  $\rightarrow nk$

$$\left[ \frac{nk \log nk}{nk \log nk} \right] = nk \log nk$$



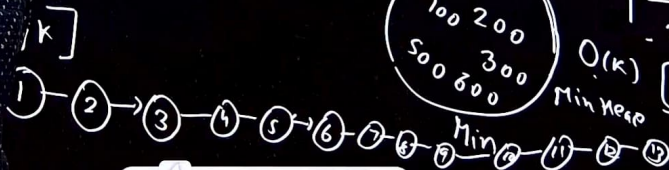
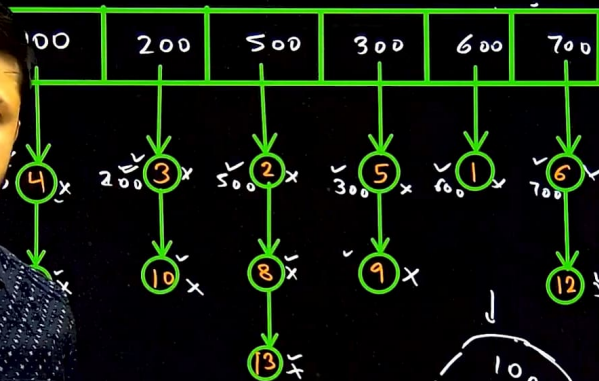
$$[nk \log k]$$





# Merge K sorted Linked List

K-elements:



$$\left( \log K + \log K + 1 \right) \Rightarrow \log K$$

$K=6$

- ① Brute force
- ② Merge-sort

$$\frac{nK \log K}{\log K}$$

③  $\Rightarrow$  Heap  $\checkmark$   
Space  $\rightarrow nK$

$$\left[ \frac{nK \log nK}{\log K} - nK \log nK \right]$$

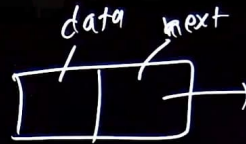
$nK \log nK$



$O(K)$   
Min Heap

class Node

```
{ public:  
    int data;  
    Node * next;  
    Node (int x)  
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      next = NULL;  
    }  
};
```

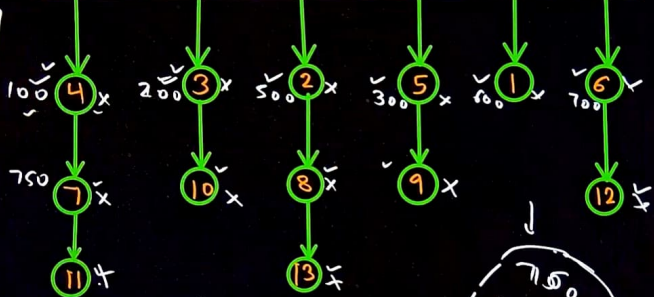


Priority\_queue < Node\*, vector<Node\*>, greater<Node\*> > p;

# Merge K sorted Linked List

Node #  
800 ✓

K-elements:



$$\frac{109K}{109K+1} \Rightarrow \log K$$

K=6

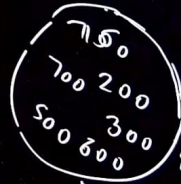
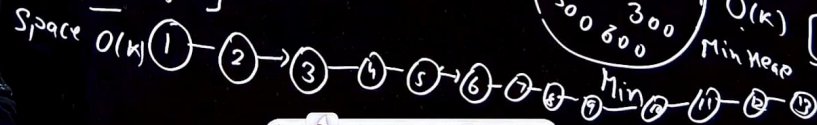
- ① Brute force
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$$\frac{nK \log K}{nK}$$

③  $\Rightarrow$  Heap ✓  
Space  $\rightarrow nK$

$$\frac{nK \log nK}{nK \log nK} = nK \log nK$$

$[nK \log K]$  ④  $\rightarrow$

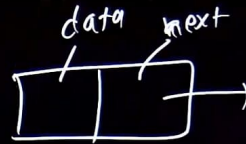


O(K)  
Min Heap



class Node

```
{ public:  
    int data;  
    Node * next;  
    Node (int x)  
    { data = x;  
      next = NULL;  
    }  
};
```



Priority\_queue < Node\*, vector<Node\*>, Compare > p;

class Compare  
{ public:

bool operator() (Node \*a, Node \*b)  
{

return a->data > b->data;

}

(Min heap)

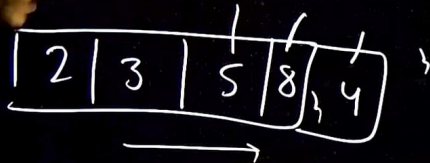


class Compare  
{ public:

bool operator() (Node \*a, Node \*b)

{  
return a->data < b->data;  
}

int  
6 9 6  
5 <= 8



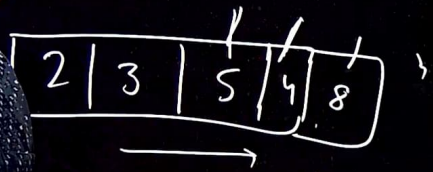
```
class Compare
{
public:
```

```
    bool operator() (Node * a, Node * b)
```

```
    {
        return a->data < b->data;
    }
```

bool ✓  
int

int  
6 9 6  
5 <= 8



8 <= 4

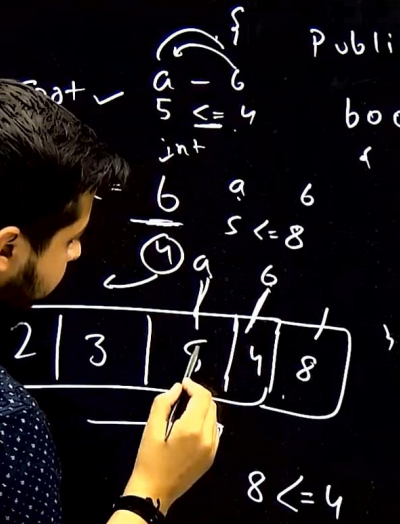


# Class Compare

Public:

bool operator() (Node \* a, Node \* b)

return a->data < b->data;



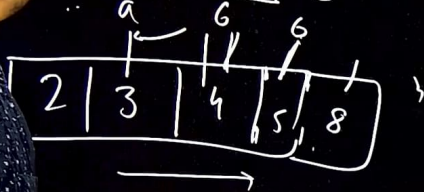
class Compare  
{  
public:

bool operator() (Node \*a, Node \*b)

{  
return a->data <= b->data;  
}

a <= b

increasing



8 <= 4

class Compare

public:

bool operator() (Node \* a, Node \* b)

return a->data <= b->data;

लेकिन यहां पर य उल्टा काम करता है हीप के  
केस में ही के केस में उल्टा काम



class Compare  
{  
public:

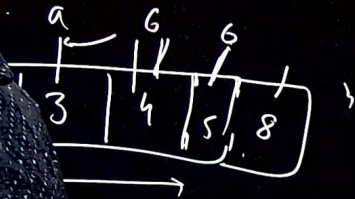
bool operator() (Node \*a, Node \*b)  
{

return a->data > b->data;

Min heap

a->data < b->data

Max heap



$8 \leq 4$



Problem

Editorial

Submissions

Comments

C++ (g++ 5.4)

Average Time: 60m

Start Timer



## Merge K sorted linked lists



Medium

Accuracy: 57.01%

Submissions: 77K+

Points: 4

Given K sorted linked lists of different sizes. The task is to merge them in such a way that after merging they will be a single sorted linked list.

Example 1:

Input:

K = 4

value = {{1,2,3},{4 5},{5 6},{7,8}}

Output: 1 2 3 4 5 5 6 7 8

Explanation:

The test case has 4 sorted linked list of size 3, 2, 2, 2

1st list 1 -&gt; 2 -&gt; 3

2nd list 4 -&gt; 5

3rd list 5 -&gt; 6

4th list 7 -&gt; 8

The merged list will be

1-&gt;2-&gt;3-&gt;4-&gt;5-&gt;5-&gt;6-&gt;7-&gt;8.

Example 2:

Input:

K = 3

```
38 Node(int x){
39     data = x;
40     next = NULL;
41 }
42
43 };
44 */
45
46 class Solution{
47 public:
48
49     //Function to merge K sorted linked list.
50     Node * mergeKLists(Node *arr[], int K)
51     {
52         // Your code here
53
54     }
55 };
56
57
58 // } Driver Code Ends
```



Custom Input

Compile &amp; Run

Submit



&lt;/&gt; Problem

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Example 2:

Input:

K = 3

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TO CLASS SOLUTION

47 public:

48

49 class Compare

50 {

51 public:

52 bool operator()(Node \*a, Node \*b)

53 {

54

55 }

56 }

57

58 //Function to merge K sorted linked list.

59 Node \* mergeKLists(Node \*arr[], int K)

60 {

61 // Your code here

62 // Min heap

63 priority\_queue&lt; Node \*, vector&lt; Node \*&gt; , Compar

64

65

66

67 }



Custom Input

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</> Problem

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Example 2:

Input:  
K = 3  
value = {{1,2},{4,5,6},{7,8}}

```

47 public:
48
49 class Compare
50 {
51     public:
52     bool operator()(Node *a, Node *b)
53     {
54         return a->data > b->data;
55     }
56 };
57
58 // a and b ka order sahi hai
59
60 //Function to merge K sorted linked list.
61 Node * mergeKLists(Node *arr[], int K)
62 {
63     // Your code here
64     // Min heap
65     priority_queue< Node *, vector< Node *> , Compare>
66
67

```



Custom Input

Compile & Run

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Problem

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56 };

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59 // Heap mein reverse tarique se kaam karta hai

60 //Function to merge K sorted linked list.

61 Node \* mergeKLists(Node \*arr[], int K)

62 {

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65 priority\_queue< Node \*, vector< Node \*> , Compar

66

67



Custom Input

Compile & Run

Submit



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4th list 7->8

आपको इनिश इज करना है ठीक हां जी तो एआर कॉमा एआर k ठीक है

Input:  
K = 3  
value = {{1,2},{4,5,6},{7,8}}

```
53
54 return a->data > b->data;
55
56
57
58 // b ka order sahi hai
59 mein reverse tarike se kaam karta hai
60 on to merge K sorted linked list.
61 ergeKLists(Node *arr[], int K)
62
63 // Your code here
64 in heap
65 rity_queue< Node *, vector< Node *> , Compare>p(arr, arr
66
67
68
69
70
71
72
73 er Code Ends
```



Custom Input

Compile & Run

Submit

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Example 2:

Input:

K = 3

value = {{1,3},{4,5,6},{8}}

Output: 1 3 4 5 6 8

```

59 // Heap mein reverse tarike se kaam karta hai
60 //Function to merge K sorted linked list.
61 Node * mergeKLists(Node *arr[], int K)
62 {
63     // Your code here
64     // Min heap
65     priority_queue< Node *, vector< Node *> , Compare
66
67     Node *root = new Node(0);
68     Node *tail = root;
69
70     Node *temp;
71
72     while(!p.empty())
73     {
74         |
75     }
76
77
78
79 }

```



&lt;/&gt; Problem

Editorial

Submissions

Comments

C++ (g++ 5.4)

Average Time: 60m

Start Timer



Output Window



Compilation Results

Custom Input

Suggest Feedback

Request Queued

Ready for evaluation



```
63 // Your code here
64 // Min heap
65 priority_queue< Node *, vector< Node *> , Compar
66
67 Node *root = new Node(0);
68 Node *tail = root;
69
70 Node *temp;
71
72 while(!p.empty())
73 {
74     temp = p.top();
75     p.pop();
76     tail->next = temp;
77     tail = tail->next;
78     if(temp->next)
79         p.push(temp->next);
80 }
81
82 return root->next;
83
84
```



Custom Input

Compile &amp; Run

Submit